Kernel theory

convergence of the iterate–test loop

- T1: A1 finite frame
- A2 local ideation
- A3 persistent generativity
- A4 unbiased repeated tests
- A5 retention
- T2 bias: $2B$-local optimum
- T3 drift: $4D$-local optimum
- T4 Pareto archive
- Optimiser's curse; σ^2/Δ^2 tests

arXiv preprint (2026)

Design principles

for an experimentation IS

- DP1 retain every tested variant
- DP2 generate near the leaders
- DP3 screen before building
- DP4 re-test leaders; size tests
- DP5 discount thin evidence
- DP6 audit persistent bias
- DP7 forget after shifts
- DP8 archive trade-offs

Gregor et al. (2020)

Artefact

evidence-ledger experimentation IS



- evidence ledger (belief + count)
- idea pool with screening scores
- re-test planner
- ship rule: lower bound $M - \kappa s$
- bias audit (A/A + holdout)
- forgetting factor γ
- Pareto archive

instantiated in the testbed

Evaluation in the testbed

hypotheses and experiments

- E1 validation of the testbed
- E2 dynamics: H1
- E3 governance policies: H2
- E4 new vs re-test: H3
- E5 GenAI ideation: H4
- E6 measurement bias: H5
- E7 drift and shift: H6
- E8 stakeholders: H7
- E9 robustness: H8

tens of thousands of simulated projects